

Bayesian multimodeling: Bayesian evidence and sampling methods

MIPT

2026

Recap

Given data $\mathbf{X}$ and parameters $\mathbf{w}$:

- Likelihood

$$p(\mathbf{X} \mid \mathbf{w})$$

- Prior

$$p(\mathbf{w} \mid \mathbf{h})$$

- Posterior

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{h}) \propto p(\mathbf{X} \mid \mathbf{w})p(\mathbf{w} \mid \mathbf{h})$$

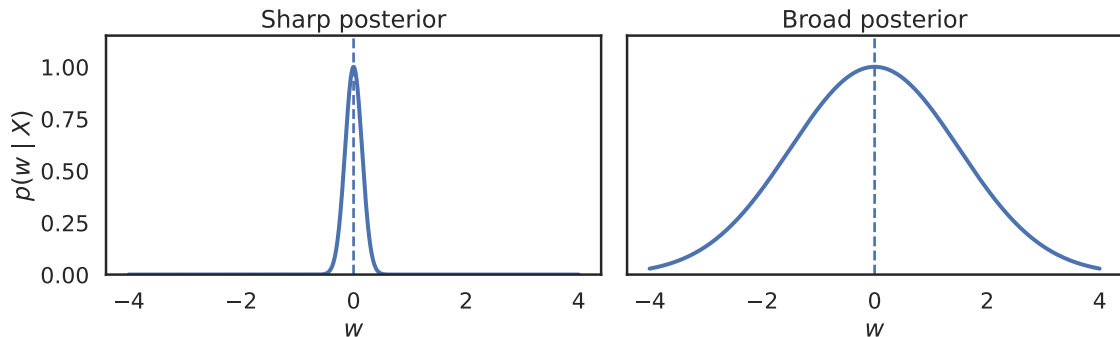
- MLE

$$\hat{\mathbf{w}}_{\text{MLE}} = \arg \max_{\mathbf{w}} p(\mathbf{X} \mid \mathbf{w})$$

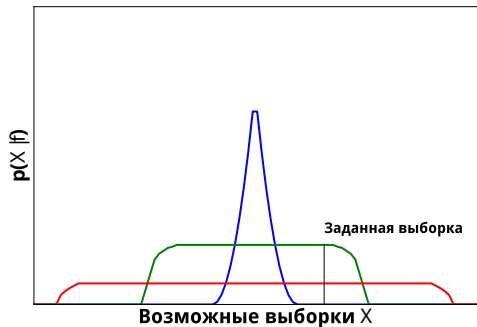
- MAP

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg \max_{\mathbf{w}} p(\mathbf{X} \mid \mathbf{w})p(\mathbf{w} \mid \mathbf{h})$$

MAP: when we cannot use a pointwise estimation



Model selection problem



- Which model should we select?
- Which is the simplest model?
- Which is the model with the best fit?

Model selection problem: Bayesian coherent inference

First level: find optimal parameters:

$$\hat{w} = \arg \max \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w}|\mathbf{h})}{p(\mathcal{D}|\mathbf{h})},$$

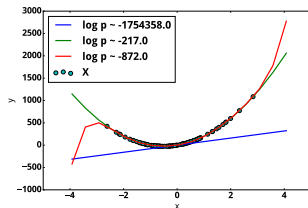
Second level: find model, that gives optimal Evidence.

“Evidence”:

$$p(\mathcal{D}|\mathbf{h}) = \int_{\mathbf{w}} p(\mathcal{D}|\mathbf{w})p(\mathbf{w}|\mathbf{h})d\mathbf{w}.$$

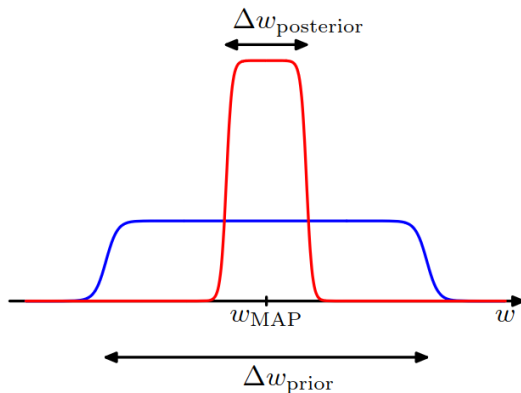


Model selection scheme



Example: polynomial regression

MAP as an approximation of Evidence



From Bishop: “we can obtain a rough approximation to the model evidence if we assume that the posterior distribution over parameters is sharply peaked around its mode w_{MAP} .”

Example: linear regression

Given m objects with n features:

$$\mathbf{f}(\mathbf{X}, \mathbf{w}) = \mathbf{X}\mathbf{w}, \quad \mathbf{y} \sim \mathcal{N}(\mathbf{f}(\mathbf{X}, \mathbf{w}), \beta^{-1}), \quad \mathbf{w} \sim \mathcal{N}(0, \mathbf{A}^{-1}).$$

The Evidence is

$$\begin{aligned} p(\mathcal{D}|\mathbf{h}) &= p(\mathbf{y}|\mathbf{X}, \mathbf{A}, \beta) \\ &= \frac{\sqrt{\beta|\mathbf{A}|}}{\sqrt{(2\pi)^{m+n}}} \int_{\mathbf{w}} \exp\left[-\frac{\beta}{2}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top(\mathbf{y} - \mathbf{X}\mathbf{w})\right] \exp\left[-\frac{1}{2}\mathbf{w}^\top \mathbf{A}\mathbf{w}\right] d\mathbf{w}. \\ &= \frac{\sqrt{\beta|\mathbf{A}|}}{\sqrt{(2\pi)^{m+n}}} \int_{\mathbf{w}} \exp(-S(\mathbf{w})) d\mathbf{w}. \end{aligned}$$

Example: linear regression

For the linear regression model, define

$$\mathbf{H} = \mathbf{A} + \beta \mathbf{X}^\top \mathbf{X}, \quad \hat{\mathbf{w}} = \beta \mathbf{H}^{-1} \mathbf{X}^\top \mathbf{y}.$$

Then

$$\int_{\mathbf{w}} \exp(-S(\mathbf{w})) d\mathbf{w} = (2\pi)^{n/2} \exp(-S(\hat{\mathbf{w}})) |\mathbf{H}^{-1}|^{1/2}.$$

Therefore,

$$p(\mathbf{y}|\mathbf{X}, \mathbf{A}, \beta) = \frac{\beta^{m/2} |\mathbf{A}|^{1/2}}{(2\pi)^{m/2} |\mathbf{H}|^{1/2}} \exp(-S(\hat{\mathbf{w}}))$$

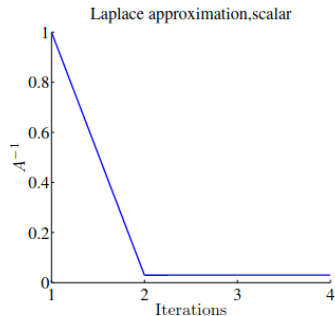
where

$$S(\hat{\mathbf{w}}) = \frac{\beta}{2} \|\mathbf{y} - \mathbf{X} \hat{\mathbf{w}}\|^2 + \frac{1}{2} \hat{\mathbf{w}}^\top \mathbf{A} \hat{\mathbf{w}}.$$

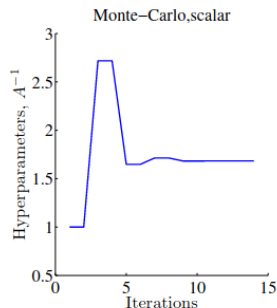
We can find the value of the Evidence for the linear models.

But what if the integral is intractable?

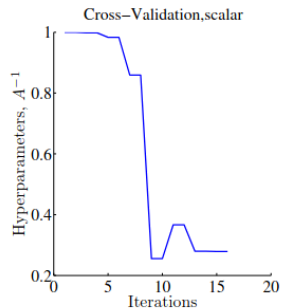
Hyperparameter selection for linear model (Kuznetsov et al.)



(a) Laplace approximation

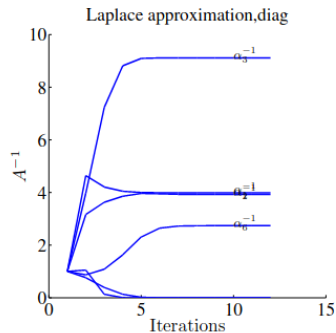


(b) Monte-Carlo

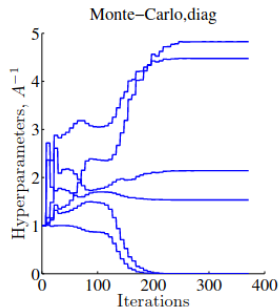


(c) Cross validation

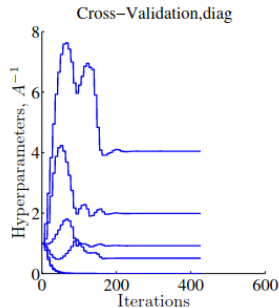
Hyperparameter selection for linear model (Kuznetsov et al.)



(a) Laplace approximation

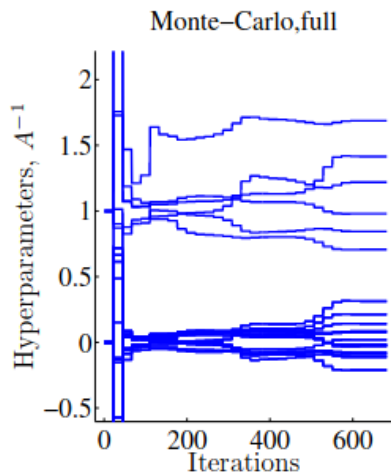


(b) Monte-Carlo

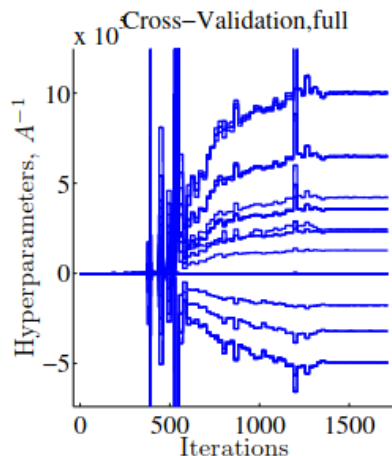


(c) Cross validation

Hyperparameter selection for linear model (Kuznetsov et al.)



(a) Monte-Carlo



(b) Cross validation

Monte Carlo sampling: naive method

$$I = \mathbb{E}_{p(\boldsymbol{w})}[f(\boldsymbol{w})] = \int f(\boldsymbol{w})p(\boldsymbol{w}) d\boldsymbol{w}.$$

If we can sample

$$\boldsymbol{w}^{(1)}, \dots, \boldsymbol{w}^{(N)} \sim p(\boldsymbol{w}),$$

then

$$\hat{I} = \frac{1}{N} \sum_{i=1}^N f\left(\boldsymbol{w}^{(i)}\right)$$

is an estimator of I .

Why does this not solve our problem?

Properties of Monte Carlo estimation

For independent samples $\boldsymbol{w}^{(i)} \sim p(\boldsymbol{w})$:

- **Strongly consistent:**

$$\hat{I} \xrightarrow{\text{a.s.}} I$$

- **Unbiased:**

$$\mathbb{E}[\hat{I}] = I$$

- **Asymptotically normal:**

$$\sqrt{N}(\hat{I} - I) \xrightarrow{d} \mathcal{N}(0, \text{Var}_p[f])$$

- **Variance:**

$$\text{Var}(\hat{I}) = \frac{\text{Var}_p[f]}{N} = O\left(\frac{1}{N}\right)$$

Challenge: we need to sample from $p(\boldsymbol{w})$.

Inverse transform sampling

Suppose $u \sim \mathcal{U}(0, 1)$ and we know the inverse CDF F^{-1} of the target distribution. Then

$$w = F^{-1}(u)$$

has CDF F .

Example: exponential distribution

$$p(w) = \lambda e^{-\lambda w}, \quad w \geq 0.$$

$$F(w) = 1 - e^{-\lambda w},$$

therefore

$$F^{-1}(u) = -\frac{1}{\lambda} \log(1 - u).$$

Rejection sampling

Suppose we know $p(w)$ up to a normalizing constant and can sample from an auxiliary distribution $q(w)$.

Choose k such that

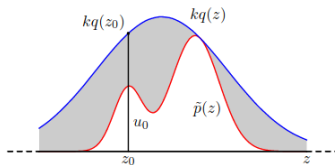
$$p(w) \leq kq(w) \quad \forall w.$$

Then:

- 1 Sample $w_0 \sim q(w)$.
- 2 Sample $u \sim \mathcal{U}(0, kq(w_0))$.
- 3 Accept w_0 if

$$u \leq p(w_0).$$

Accepted samples follow $p(w)$.



Importance sampling

Suppose we cannot sample from $p(w)$ directly, but can evaluate $p(w)$.
Let $q(w)$ be an auxiliary distribution from which we can sample:

$$\begin{aligned}\mathbb{E}_p[f(w)] &= \int f(w)p(w) dw \\ &= \int f(w)\frac{p(w)}{q(w)}q(w) dw \\ &= \mathbb{E}_q\left[f(w)\frac{p(w)}{q(w)}\right].\end{aligned}$$

Therefore, with $w^{(l)} \sim q$,

$$\hat{I} = \frac{1}{L} \sum_{l=1}^L f(w^{(l)}) \frac{p(w^{(l)})}{q(w^{(l)})}.$$

Markov Chain Monte Carlo

Similar to rejection sampling, MCMC uses an accept/reject mechanism to construct a Markov chain whose stationary distribution is the target distribution $p(w)$.

Basic idea:

Construct a Markov chain whose stationary distribution is the target distribution $p(w)$.

$$w^{(0)} \rightarrow w^{(1)} \rightarrow w^{(2)} \rightarrow \dots$$

We want

$$p(w) = \lim_{t \rightarrow \infty} p(w^{(t)}).$$

A sufficient condition is **detailed balance**:

$$p(w')T(w|w') = p(w)T(w'|w).$$

Metropolis–Hastings algorithm

- Sample a proposal $w' \sim q(w'|w^t)$.
- Accept with probability

$$A(w'|w^t) = \min \left(1, \frac{p(w')q(w^t|w')}{p(w^t)q(w'|w^t)} \right).$$

- If accepted: $w^{t+1} = w'$.
- Otherwise: $w^{t+1} = w^t$.

The transition probability for $w' \neq w$ is

$$T(w'|w) = q(w'|w)A(w'|w).$$

Then

$$\begin{aligned} p(w)T(w'|w) &= p(w)q(w'|w) \min \left(1, \frac{p(w')q(w|w')}{p(w)q(w'|w)} \right) \\ &= \min (p(w)q(w'|w), p(w')q(w|w')) \\ &= p(w')T(w|w'). \end{aligned}$$

Thus, the detailed balance condition is satisfied:

$$\boxed{p(w)T(w'|w) = p(w')T(w|w')}$$

Gibbs sampling

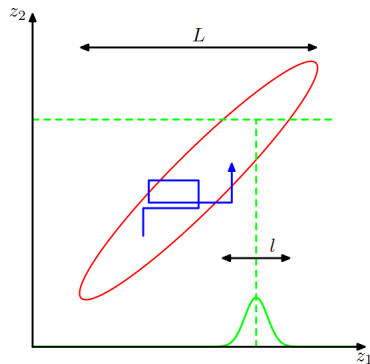
Given n variables $w_1, \dots, w_n$, we want to sample from their joint distribution $p(\mathbf{w})$. The idea is simple: sample variables sequentially from their conditional distributions. Starting from an arbitrary $\mathbf{w}^{(0)}$:

$$\begin{aligned}w_1^{(t+1)} &\sim p(w_1 \mid w_2^{(t)}, \dots, w_n^{(t)}), \\w_2^{(t+1)} &\sim p(w_2 \mid w_1^{(t+1)}, w_3^{(t)}, \dots, w_n^{(t)}), \\&\vdots \\w_n^{(t+1)} &\sim p(w_n \mid w_1^{(t+1)}, \dots, w_{n-1}^{(t+1)}).\end{aligned}$$

After a burn-in period, $\mathbf{w}^{(t)}$ can be used as samples from the joint distribution.

Gibbs sampling: properties

- Can we consider Gibbs sampling as a special case of Metropolis–Hastings?
- The efficiency strongly depends on the correlations between the variables.

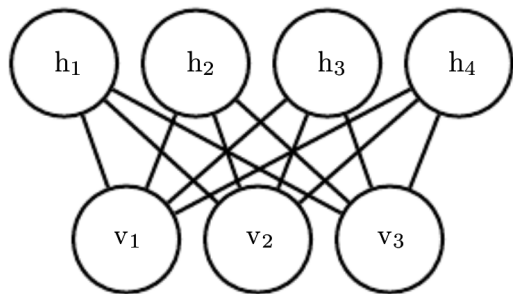


From Bishop: the step size is governed by the standard deviation of the conditional distribution (green curve), and is $O(l)$. For an elongated joint distribution (red ellipse), the number of steps needed to obtain an approximately independent sample is $O\left(\frac{L^2}{l^2}\right)$.

Restricted Boltzmann machine

An early deep learning architecture:

$$E(\mathbf{x}, \mathbf{h}) = -\mathbf{h}^\top \mathbf{W} \mathbf{x} - \mathbf{w}_{bh}^\top \mathbf{h} - \mathbf{w}_{bb}^\top \mathbf{x}.$$



Contrastive Divergence: idea

Energy-based model:

$$p(x|w) = \frac{\exp(-E_w(x))}{Z(w)}, \quad Z(w) = \int_x \exp(-E_w(x)) dx.$$

The gradient of the log-likelihood is

$$\frac{\partial \log p(x|w)}{\partial w} = \mathbb{E}_{x' \sim p(x|w)} \left[\frac{\partial E(x')}{\partial w} \right] - \frac{\partial E(x)}{\partial w}.$$

For an RBM, samples from $p(x|w)$ can be obtained using Gibbs sampling:

- Take x_0 from the dataset
- $h_0 \sim p(h|x_0)$
- $x_1 \sim p(x|h_0)$
- $h_1 \sim p(h|x_1)$
- ...
- Obtain x_k

$$\frac{\partial [-\log p(x_0|w)]}{\partial w} \approx \frac{\partial E(x_0)}{\partial w} - \frac{\partial E(x_k)}{\partial w}.$$

Optimization of q

The efficiency of Metropolis–Hastings strongly depends on the proposal distribution $q(w'|w)$. A simple choice is a Gaussian random walk:

$$q(w'|w) = \mathcal{N}(w'; w, \sigma^2 I).$$

More expressive proposals can be parameterized by neural networks.

- Normalizing flows can parameterize expressive proposal distributions.
- Invertibility allows us to evaluate the proposal density using the change-of-variables formula.

Optimization variants:

- Entropy $\times$ acceptance rate (Li et al., 2020)
- GAN between empirical distribution and q (Song et al., 2017)

Hamiltonian (aka Hybrid) Monte Carlo¹

Consider an energy-based model:

$$p(\mathbf{w}) = \frac{\exp(-E(\mathbf{w}))}{Z},$$

where Z is a normalizing constant (possibly unknown).

Main idea: introduce a momentum $\mathbf{m}$ and kinetic energy

$$K(\mathbf{m}) = \frac{1}{2} \mathbf{m}^\top \mathbf{m}.$$

Then define the Hamiltonian

$$H(\mathbf{w}, \mathbf{m}) = E(\mathbf{w}) + K(\mathbf{m}),$$

and the joint distribution

$$p_H(\mathbf{w}, \mathbf{m}) = \frac{1}{Z_H} \exp(-H(\mathbf{w}, \mathbf{m})).$$

Since the Hamiltonian is separable,

$$p_H(\mathbf{w}, \mathbf{m}) = p(\mathbf{w})p(\mathbf{m}),$$

so the marginal distribution of $\mathbf{w}$ is exactly the target distribution $p(\mathbf{w})$.

¹see the tutorial talk of Denis Rubtsov, 2025

Hamiltonian dynamics

$$\frac{\partial w_i}{\partial t} = \frac{\partial H}{\partial m_i} = m_i, \quad \frac{\partial m_i}{\partial t} = -\frac{\partial H}{\partial w_i} = -\frac{\partial E}{\partial w_i}.$$

- During the evolution of this dynamical system, the Hamiltonian H is constant.
- Liouville's theorem: the volume of the phase space induced by the dynamics is constant.
- **Therefore:** Hamiltonian dynamics preserves the joint distribution $p_H(\mathbf{w}, \mathbf{m})$.
- **However,** the dynamics is not necessarily ergodic: a trajectory remains on a surface of constant H .
- To obtain an ergodic sampling scheme, we need additional moves that change H while preserving p_H .

HMC proposals

There are multiple ways to sample $\mathbf{m}$ for HMC:

- **Naive (see MacKay book):** draw $\mathbf{m}$ from Gaussian density using Gibbs sampling. The proposal is always accepted.
- **Dynamical:**

$$\frac{\partial \mathbf{w}}{\partial t} = \mathbf{m}, \quad \frac{\partial \mathbf{m}}{\partial t} = -\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}}.$$

- ▶ The proposals are accepted by the MH rule.
- ▶ The proposals are always accepted if the simulation is perfect. In reality some samples can be discarded due to the finite integration steps, for example.
- ▶ The state of the system tends to move a distance that goes linearly with the computer time.

Practical issue: we need to integrate over time to perform an HMC step. The standard way to do this is the **leapfrog** integrator, which preserves the volume up to a small numerical error.

HMC: limitations and comparison to MH ²

- De-facto standard for accurate sampling in low- to medium-dimensional continuous parameter spaces; HMC is often used as a reference method in the literature
- Usually a small rejection rate
- Usually faster than MH, but requires careful hyperparameter tuning
- Still impractical for posterior sampling during training in large, high-dimensional parameter spaces

²https://nbviewer.org/github/bynchang/HMC-Tutorial/blob/master/hmc_tutorial.html

What are Bayesian neural network posteriors really like?

Izmailov et al., 2021:³

- HMC for posterior distribution estimation for deep models on some standard datasets.
- Resources: 512 TPU

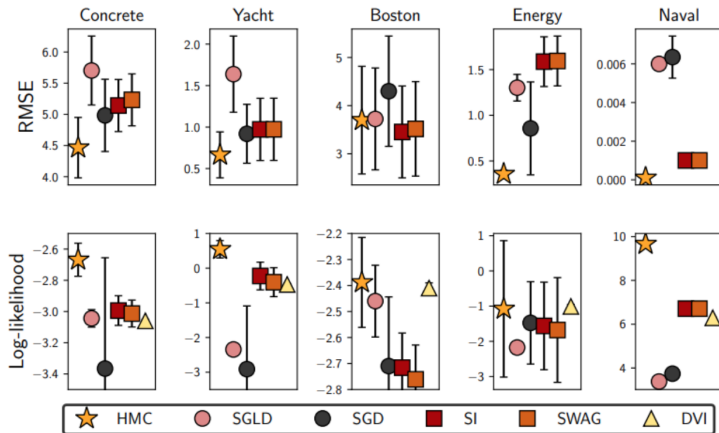
Some conclusions:

- HMC trajectories: longer HMC trajectories allow the chain to explore the posterior more efficiently.
- HMC step sizes: decreasing step size leads to higher accept rate.
- The posterior contains connected high-density regions that are functionally diverse and explorable by HMC.
- Cold posteriors are not required for good results and in fact can hurt performance.
- HMC BNNs are fairly robust to Gaussian prior variance.

³https://docs.google.com/presentation/d/1WPjqKw3b-TpPSaHcwhqFsuAE575_nDigit5SVoP3CoNI/edit?usp=sharing

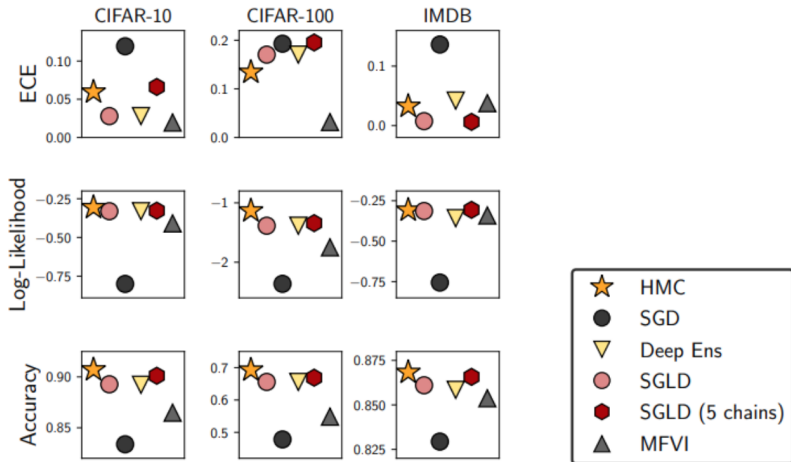
What are Bayesian neural network posteriors really like?

BNN evaluation: UCI



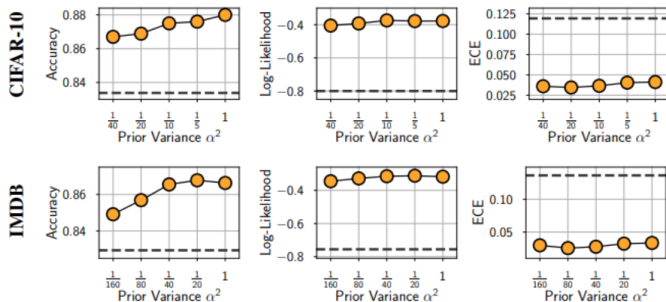
What are Bayesian neural network posteriors really like?

BNN evaluation: CIFAR and IMDB



What are Bayesian neural network posteriors really like?

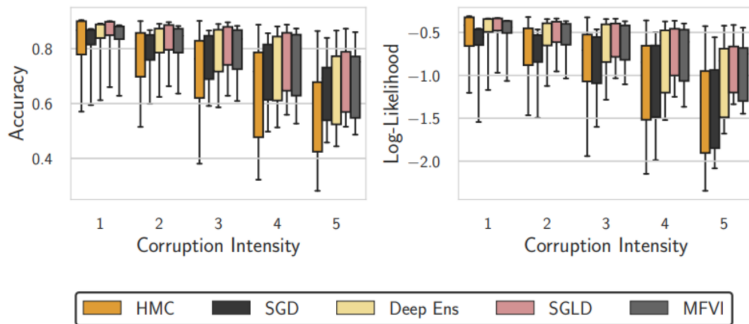
Effect of priors



HMC BNNs are fairly robust to Gaussian prior variance.

What are Bayesian neural network posteriors really like?

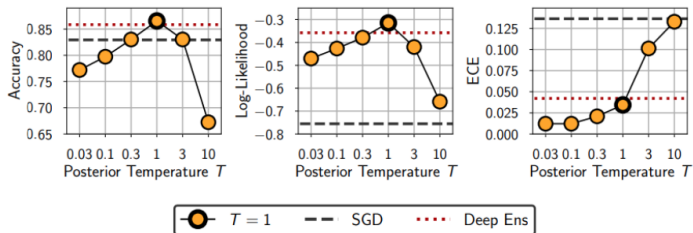
Train on CIFAR-10, test on CIFAR-10-C



What are Bayesian neural network posteriors really like?

Posterior temperature effect

- We have already seen that BNNs can do well at $T=1$
- What is the effect of T then?



Cold posteriors are not required for good results and in fact can hurt performance!

References

- Bishop C. M. Pattern recognition //Machine learning. – 2006. – T. 128. – №. 9.
- MacKay D. J. C., Mac Kay D. J. C. Information theory, inference and learning algorithms. – Cambridge university press, 2003.
- Kuznetsov M., Tokmakova A., Strijov V. Analytic and stochastic methods of structure parameter estimation //Informatica. – 2016. – T. 27. – №. 3. – C. 607-624.
- Li Z., Chen Y., Sommer F. T. A neural network mcmc sampler that maximizes proposal entropy //arXiv preprint arXiv:2010.03587. – 2020.
- Song J., Zhao S., Ermon S. A-nice-mc: Adversarial training for mcmc //Advances in Neural Information Processing Systems. – 2017. – T. 30.
- Izmailov, Pavel, et al. "What are Bayesian neural network posteriors really like?."International conference on machine learning. PMLR, 2021.

Organizational issues

No talks?

Project distribution

- Low-variance gradient estimation (choose between arbitrary distribution and mixtures)
 - ▶ Горбунова, Иванов, Пономарев, Уденеев
 - ▶ Check “Systems and means of informatics” journal
- Epiplexity
 - ▶ Иконников, Казачков, Кравацкий, Черноусов
 - ▶ Need to discuss with Andrey? Next week?
- Complexity
 - ▶ Алябушев, Дементьев, Курдюков, Чирков*

Hometask: roles

Each team must assign roles for all teammates. Each role is evaluated independently, thus number of activities per each teammate must be $\approx$ equal.

- Project planning
- Proof-of-concept (wrap baseline algorithm into library structure and present how it works)
- Algorithm implementation (**1 activity per teammate**)
- Project wrapping
- Tests writing
- Documentation writing
- Cross-review (**1 activity per teammate**)
- Final demo
- Blog post
- Tech. report

All the roles must be finished according to the project schedule (see course page). For the deadline violation the score will be multiplied by 0.66 for each week.

Next homework: presentation

- Assign roles and set them into table.
- For all teams and their members: make presentations of your projects
- The presentation must cover
 - ▶ Project description (maybe more detailed than I gave to you)
 - ▶ Name of the project library
 - ▶ Scheme of the project (what will be the classes, how it will be integrated, what's the stack)
 - ▶ Brief algorithm description (from 1 to 4 slides for all the algorithms, other people must be able to understand the idea of all the algorithms)
 - ▶ Idea for demo/basic code
- Time limit: 15 min

Next hometask: for people who are wrapping the library

- Create a repository in intsystems
- Keep in mind the future library must support documentation deploy and auto-testing. You can make the repository by yourself OR
 - ▶ Use Andrey Grabovoy's template: <https://github.com/intsystems/ProjectTemplate>
 - ★ Please turn off autodeploy of github pages
 - ▶ Use my template: <https://github.com/intsystems/SoftwareTemplate-simplified>
 - ▶ Use cookiecutter <https://cookiecutter-data-science.drivendata.org/>
 - ▶ Use any other template, see for example:
https://github.com/LauzHack/pytorch_project_template
- Think about stack:
 - ▶ Codestyle (linters?)
 - ▶ Documentation engines (mkdocs? shpinx?)
 - ▶ Test libraries (built-in unittest? pytest?)
- Please make it w.r.t. to the manual

Repository info

Before creating new repository, please read [this friendly manual](#)

Please make it w.r.t. to the manual

Repository info

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Please make it w.r.t. to the manual

Next hometask: for people who are planning the library

- Create a document in the repository with the following information (the same as in the presentation, but maybe with more details)
 - ▶ Project name
 - ▶ Architecture of the project: what classes must be implemented? How they should interact?
 - ▶ Describe all the public functions and class methods you are planning to implement, with annotations.
 - ▶ What are the libraries you are planning to use and/or integrate?
- In perfect case, the member who is implementing the algorithm can write the code just by your architecture description.
- Note, the document can be improved/changed in the future, but I will score you and other members of the team on the correspondence of the proposed structure and the final algorithm implementation.

For other activities

- I will add some comments on the next meeting for all the activities.
- PoC code: deadline is 20th of October. You can start thinking together with demo code.
- Blog post and tech. report: very drafty version must be ready on the 20th of October. Have a look at the posts of the previous years.
- Documentation: structure of the documentation must be ready on the 20th of October. Discuss the engine of the documentation with a person responsible for the repo.
- Tests: must be ready on the 24th of November with some high coverage (70%?). Discuss the framework of the tests with a person responsible for the repo.
- Algorithms: must be ready on the 24th of November.